

Releasing a Feature Channel Beside a Prediction Service

You Cannot Change:

An Evaluated Release Problem and Six Negative Mechanism Results

Anonymous Submission

Abstract—An organisation already publishes a prediction service: fixed probability vectors that downstream recipients consume and cannot be asked to re-accept. We formalise and evaluate what happens when it adds a second, reusable feature channel for one recipient without altering a published number, while limiting what that recipient, and that recipient colluding with another, can newly infer about sex and race. Our contribution is an implemented release interface and measurement contract: per-recipient and coalition views, independently fitted attackers, incremental disclosure reported beside absolute disclosure, and negative increments retained unclipped. On a locked, previously unused survey year, a penalty built from the coalition’s joint view beats an equal-strength and an equal-total-mass local control on 14 of 16 sensitive cells, none worse; its utility cost is bounded by a point rule, not by an interval. Six subsequent mechanisms fail to improve on the strongest channel we already had, under both registered senses of competitiveness: lower disclosure at comparable utility, and higher utility at bounded disclosure. Adapted published erasers beat our own mechanism. A capacity diagnostic then shows why the most recent attempt could not have succeeded as designed: a label-free code of the permitted inputs carries genuine signal for the held-out task, yet adds nothing measurable once the already-released service predictions are in the view. We report the release problem, the evaluation design, and every claim this revision withdraws.

I. INTRODUCTION

A company runs two prediction services. One returns credit-style probabilities to a business partner; the other returns coverage probabilities to a different partner. Both partners have built systems on those exact numbers, so the company cannot revise them. It now wants to give the first partner something more useful—a compact reusable feature vector built from what the published predictions leave unexplained—while making sure that the partner, and the two partners pooling what they hold, cannot use it to work out an applicant’s sex or race any better than the published predictions already allow.

That situation is ordinary and it is not what the fairness and concept-erasure literature usually evaluates. Most methods assume the releaser owns the representation and may edit or replace it. Here the published output is immutable: it is already in the recipients’ systems, and the only decision left is what to put *beside* it. That constraint changes what a mechanism may do, what preservation means, and who the adversary is.

This paper contributes the problem, an implemented interface for it, a measurement contract, and the results of six mechanisms evaluated through them. One effect replicates on a sealed year. Every mechanism we built to turn that effect into a competitive release failed, and three adapted published erasers beat our own mechanism on the same wire. We report that plainly, because the evaluation design is what makes the failures legible, and because the negative results are specific enough to constrain what should be tried next.

Contributions:

- **A release problem with an immutable published output**, formalised with per-recipient and coalition audit views, and an interface in which preservation is a bitwise identity rather than a bounded perturbation (§II).
- **A measurement contract** that makes heterogeneous mechanisms comparable at matched width on one wire: independently fitted attacker slates, incremental disclosure always reported beside absolute disclosure, negative increments retained unclipped, per-endpoint results never averaged (§II).
- **One confirmed effect** on a locked, previously unused survey year: coalition-conditioned penalties beat both matched local controls (§V).
- **Six negative mechanism results** against the strongest channel we had, under both registered senses of competitiveness, with adapted published erasers outperforming our mechanism (§VI).
- **A capacity diagnostic** explaining why the most recent design could not have worked as specified, and a delimitation of exactly what it rejects (§VII).

What this paper is not. It does not present a deployable competitive mechanism; we do not have one. It offers no new mathematics: concept erasure, eigenvalue-sign rank selection, minimax adversarial training and conditional-moment characterisations are prior work, cited as such, and adversarially protected transferable representations are not new either [1]. The framework does not establish a superior method, and we do not claim that it does.

II. THE INTERFACE AND THE MEASUREMENT CONTRACT

Each person has covariates X , sensitive attributes S (sex, 2 classes; race, 9 classes) and task labels. Two services are frozen and published before anything else happens: $H_A(X) \in [0, 1]^4$, the income-over-\$50k and employment probability vectors released to recipient A, and $H_B(X) \in [0, 1]^2$, the coverage vector released to B. A mechanism may add a channel $Z \in \mathbb{R}^{16}$ for A only, so A sees (H_A, Z) , B sees H_B , and the coalition AB sees (H_A, Z, H_B) . H without a subscript is the no-channel interface. The policy authorises A for income, employment and residence, and B for coverage and commute; sex and race are forbidden everywhere, along with nine cross-role combinations. The adversary holds labelled data from the release year, fits fixed attack families on its own pool, selects on a validation pool, and sees only its legal view. Inference for A uses A-side inputs only; H_B enters fitting and auditing and is never an inference input.

Three names recur. A_0 is an unprotected neural channel. J is an adversarially trained 16-dimensional channel from earlier work in this line, never retrained; it is the strongest comparator here, and it is *internal*—an experimental artifact of this project, not a deployed product (§VIII). $L1$ and $L2$ are local controls whose penalty uses only A’s own view, matched to the candidate on penalty strength and on total penalty mass respectively; $C1$ adds a penalty term built from the coalition’s joint view.

Four distinctions carry the argument and are never merged.

- **Preservation is architectural; accuracy is empirical.** Z is concatenated to H_A and never mixed into it, so H is preserved by construction and checked bitwise on every released view. It does *not* follow that the service stays accurate: on identical released vectors, employment log loss rises by 0.0138 nats and income falls by 0.0145 between two survey years.
- **Added capability is measured against the correct H -only view.** A channel adds capability only if independently fitted probes do better with (H_A, Z) than with H_A alone.
- **Added disclosure is a finite-auditor log-loss gain, not an information bound.** Every recovery number is the improvement a validation-selected attacker achieves over the same family restricted to H . It is a floor on what is recoverable, never a ceiling.
- **Recipients have different permitted tasks**, so the coalition’s authorised union is not either singleton’s and the audit roles differ accordingly.

Scope limits. Only recipient A receives a channel, so nothing here is a multi-channel design. Residence and commute labels are excluded from every representation fit and selection, but residence outcomes have been inspected repeatedly across this line of work, so residence is *reserved from training* rather than a pristine unseen task, and we never describe a result on it as blind evaluation.

Data and uncertainty. California ACS one-year PUMS, ages 19–34, three household splits, with disjoint pools for

representation fitting, source validation, task fitting and validation, attacker fitting and validation, and evaluation. Attacks are logistic regression, restarted MLPs, gradient-boosted trees and a random-feature kernel family, plus every legal projection of H ’s own attacks into each view. Uncertainty is a paired household-cluster bootstrap with simultaneous intervals inside pre-declared families. Both weightings (unweighted and person-weighted) are computed from the same predictions and reported separately.

III. WHAT IS PROVED, AND OVER WHAT

Four statements are elementary; nothing else in this paper is a theorem.

- 1) Appending Z without modifying H preserves H exactly. This is a property of concatenation.
- 2) **Inclusion fact.** A recipient holding $[H, Z]$ can ignore Z , so its optimal *population* log loss is no worse than with H alone — for the task *and* for a sensitive attribute. A deterministic appended channel therefore cannot reduce optimal sensitive recovery, and a constant appended channel has exactly zero population incremental disclosure. Any reading of “appending this channel protected us” is wrong by construction; only finite fitted probes can move either way, and some of our measured increments are negative.
- 3) Under Bayes prediction and log loss, the population risk difference between H and (H, Z) equals $I(S; Z | H)$. Our validation-selected finite attackers do not attain those infima, so reported differences bound that quantity in neither direction, and mutual-information estimation has its own formal limits [2].
- 4) For one refinement, orthogonal reparameterisation leaves the objective invariant, measured to 5.6×10^{-17} .

Ky Fan’s principle [3] gives a restricted optimum for the original fixed-matrix family only; the refinements have no fixed matrix and inherit no optimality statement. The moment condition we use is a finite fitted subset of the L^2 characterisation of conditional independence behind kernel CI tests [4], [5]. Vanishing fitted moments establish no conditional independence: one of our own fixtures has a misspecified nuisance manufacturing a penalty more than ten times the oracle value under *exact* conditional independence.

IV. RELATED WORK

The closed-form line of Sadeghi et al. solves this shape of problem when the releaser owns the representation: SARL [6] takes eigenvectors of a utility-minus- λ -dependence matrix, OptNet-ARL [7] embeds closed-form players in a learned encoder, K-TOpt [8] adds random features and evaluates on Folktables [9], and U-FaTE [10] adds a conditional penalty. Our marginal arms turn out to *be* SARL-style constructions on whitened residuals, verified by bitwise rebuild. Adversarial fair and transferable representation learning is Madras et al. [1]. Linear concept erasure [11], [12] gives an exact guarantee against *linear* adversaries on a representation the

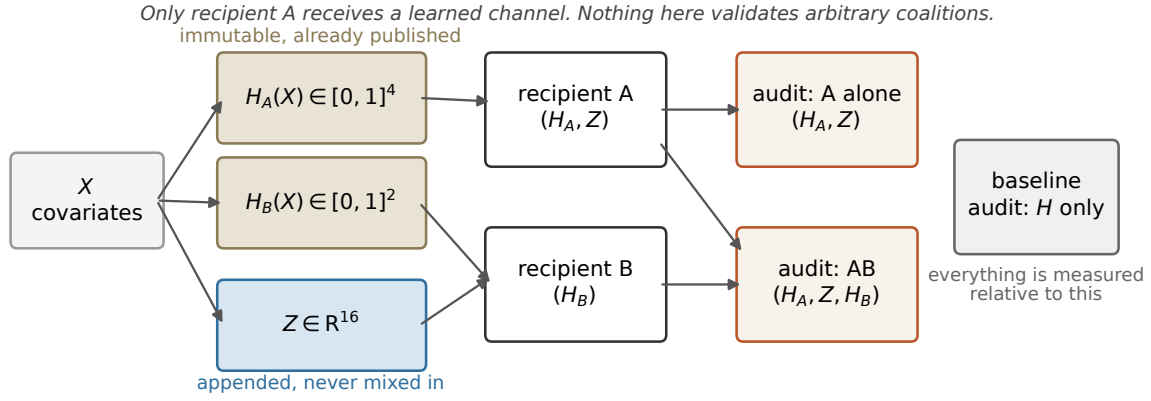


Fig. 1. The release graph. H_A and H_B are published and never modified; Z is appended to A’s wire alone. Three audit views are scored: A alone, the coalition AB, and the H -only baseline that every increment is measured against. A schematic, not a result.

releaser may edit. Adversarial removal is known to leave recoverable information [13], and least-privilege learning has fundamental limits [14]. **Reusing a known solver, a conditional-moment characterisation or a rank rule is attribution, not innovation.**

Three things differ here, with their weight stated honestly. *What we condition on*: a released continuous artifact H , through cross-fitted nuisance residuals, rather than stratification on a label. *What utility means*: reconstruction of a residualised teacher rather than dependence with a label. *Recipient structure*: separate local and coalition penalties over a fixed prior output, with only one recipient receiving a channel. The first two are adaptations of known components. For the third, the nearest prior candidate we found is the multi-consumer information-theoretic line [15], which designs one sanitised release consumed by many users; we could not confirm from its text that it carries a per-recipient release or a collusion term, and we mark that entry ambiguous rather than assert it. What we did not find, after a bounded search, is a prior *evaluation* of recipient-specific and combined-access policies over an already-published output the mechanism may not touch. That is absence of evidence, not a priority claim.

V. THE CONFIRMED EFFECT

On a locked survey year that no earlier stage of this project had touched, the coalition-conditioned candidate $C1$ has a simultaneous-interval advantage over both $L1$ and $L2$ under both weightings. Of the 16 sensitive cells, **14 have adjusted intervals entirely below zero, none is above zero, and two are unresolved** ($C1$ versus $L1$ on A/sex unweighted, $[-0.00585, +0.00020]$; versus $L2$, $[-0.00568, +0.00052]$). Frozen attacks from the earlier year reproduce the ordering, so this is not an artifact of refitting attackers on the new year.

Two qualifications travel with that result and are not separable from it. First, **the paired utility rule is a point criterion and only a point criterion**: as implemented it compares seed means at a 0.001-nat margin and passes, but **zero of four comparisons support interval non-inferiority**

at that margin, missing it by 2% to 35% even under the most permissive procedure we computed. “No statistically detected cost” and “a cost bounded below 0.001” are different statements and only the first is supported. Second, **the mechanism is not competitive with J** : $C1$ is significantly worse than J on all four sensitive endpoints under both weightings, and better only on residence capability. Since J carries materially less residence capability than the spectral arms, neither channel dominates the other.

VI. SIX MECHANISMS THAT DID NOT IMPROVE ON J

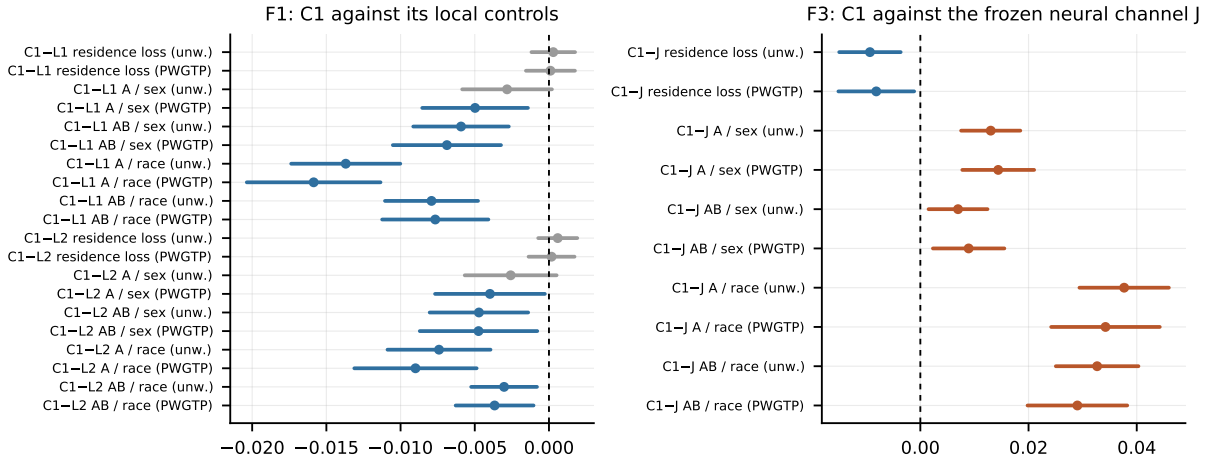
We group the negative results by what they tested rather than by the order they ran; the chronology, with evidence commits, is in Appendix A.

A. Refining the penalty

A penalty acting on nonlinear functions of the channel moved recovery in the intended direction but failed all eight registered coordination cells, and at reduced rank it made one endpoint significantly *worse* ($+0.00862 [+0.00168, +0.01556]$). That penalty was then diagnosed, after its maps had been evaluated, as coordinate-dependent: it summed degree-two monomials with equal weight instead of computing a Frobenius norm. A successor objective repaired that exactly—rotation-only share 1.4×10^{-15} over 18 cells, verified before any score was read—and **the repaired releases disclosed more**: 8 of 48 sensitive cells significantly worse, none better. The repair changed five ingredients at once, so no single cause is isolated, and we do not claim that rotation invariance is harmful.

B. Published erasers, adapted, beat our mechanism

LEACE [11] and SPLINCE [12] were applied to the auxiliary channel alone and appended to unchanged H_A ; they are deliberately *not* applied to the concatenated wire, since an oblique projection over all coordinates would generally change the published service output. Adapted LEACE and two OptNet-ARL policies recover significantly less than our repaired mechanism on all four sensitive endpoints under both



paired difference (nats); left of zero favours the candidate. Blue: interval entirely below zero. Orange: entirely above. Grey: crosses zero (unresolved, not unaffected).

Fig. 2. Paired differences with simultaneous 95% intervals inside each pre-declared family, fresh attacks on the locked year, both weightings, candidate minus comparator in nats. **Left:** the coalition-conditioned candidate against its two local controls. The two grey *A/sex* rows are *unresolved*—the comparison is undecided, not the endpoint unaffected. **Right:** the same candidate against the frozen channel *J*.

weightings, and adapted SPLINCE on seven of eight cells (Table I). **A closed-form linear eraser from 2023, applied to a channel this project already had, beats the mechanism this line of work developed across three studies.** The utility side must be stated with equal care: LEACE’s residence cost against the repaired arm is *unresolved*, $+0.0036 [-0.0028, +0.0100]$, and against the channel it actually transformed it is significant, $+0.0098 [+0.0036, +0.0159]$. Linear erasure of this channel buys its protection with measurable authorised capability, so we do not say it dominates.

C. Training the channel directly against attackers

Direct adversarial refinement of the channel, starting from the unprotected channel A_0 , produced 60 distinct new channels and improved no sensitive endpoint against any comparator: across its frontier family 189 cells are significantly worse and 18 significantly better, *all 18 on residence utility*. Its one positive—that coalition conditioning beats a local control—survives robustly only against the *weak* control; against the strength-matched control it holds on one endpoint at one width at the family-adjusted level and not under the simultaneous correction its negative results are held to. A separate finding from the same study is worth more than its headline: 63 of 114 trajectories published a checkpoint bitwise identical to their starting channel although every one had moved, so unchanged releases were a *selection* outcome, not stasis.

D. Starting from *J*, and projecting out of *J*

The most direct test initialised adversarial training from *J*’s own trained weights and, separately, removed coalition-conditioned directions from *J* by a rank-reducing projection. **No prospectively selected release met the registered competitive conjunction** under either weighting. The neural nominees are *J*: at low teacher strength the registered equal-budget selection returned the untouched starting channel on every anchor,

and at higher teacher strength the channel was pulled back toward A_0 and leaked more. The projection nominees lower coalition sex recovery against *J* ($-0.0091 [-0.0158, -0.0025]$) at a significant residence cost ($+0.0101 [+0.0023, +0.0179]$ unweighted), which is a trade-off rather than dominance, and they do not separate from matched local-only projections on any endpoint—so coalition conditioning established no advantage here. Their point estimates sit within ± 0.0011 of ordinary LEACE applied to *J* on every endpoint unweighted, which is point proximity and not statistical equivalence.

E. Buying capability instead of lowering disclosure

A release beside *H* can be competitive in two ways: lower disclosure at comparable utility, or higher utility at bounded disclosure. The studies above registered the first. A sixth registered the second: append a small learned channel *R* beside a frozen *J*, predicting residual teacher information a frozen decoder from (H_A, Z_J) leaves unexplained, and ask for capability at bounded added disclosure. Its pilot passed the capability leg and the source allowances at every configuration and **failed the declared 0.001-nat disclosure screen at every configuration**; capability and disclosure fell together, monotonically, as the adversarial weight rose (Fig. 3). The closest configuration missed the screen at the lowest capability in the grid. Its expansion tiers were never triggered, and we report them as such rather than as missing evidence.

Label-free is not neutral. The capability proxy uses no residential label, but it is the residual of A_0 , the unprotected channel that carries the largest sensitive recovery in this paper ($+0.032$ on *A/sex* and $+0.051$ on *A/race* over *H*). Rewarding reconstruction of that residual can reward sensitive-correlated directions, so “the objective itself paid for sensitive information” remains a live explanation of the pilot’s inseparability, alongside

Arm	wt.	additional recovery vs. riv16-C1				residence	source
		A / sex	AB / sex	A / race	AB / race	loss diff.	allow.
LEACE	unw.	−0.0081▼	−0.0086▼	−0.0131▼	−0.0158▼	+0.0036	3/3
	PWGTP	−0.0097▼	−0.0093▼	−0.0154▼	−0.0166▼	+0.0040	
SPLINCE	unw.	−0.0088▼	−0.0077▼	−0.0183▼	−0.0165▼	+0.0089▲	3/3
	PWGTP	−0.0100▼	−0.0084	−0.0201▼	−0.0163▼	+0.0091▲	
OptNet-C1	unw.	−0.0074▼	−0.0110▼	−0.0161▼	−0.0145▼	+0.0086▲	2/3
	PWGTP	−0.0089▼	−0.0117▼	−0.0183▼	−0.0151▼	+0.0076▲	
OptNet-L1	unw.	−0.0032	−0.0048	−0.0056	−0.0060	+0.0057	2/3
	PWGTP	−0.0059	−0.0069	−0.0116	−0.0075	+0.0046	
OptNet-L2	unw.	−0.0104▼	−0.0092▼	−0.0184▼	−0.0198▼	+0.0096▲	2/3
	PWGTP	−0.0125▼	−0.0112▼	−0.0210▼	−0.0212▼	+0.0082▲	

TABLE I

ADAPTED PUBLISHED METHODS AGAINST THE REPAIRED MECHANISM AT RANK 16 WITH THE COALITION POLICY. EVERY ARM TRANSFORMS ONLY THE AUXILIARY CHANNEL AND APPENDS IT TO UNCHANGED H_A . DEVELOPMENT POOLS. ▼ ADJUSTED SIMULTANEOUS INTERVAL ENTIRELY BELOW ZERO (THE ARM RECOVERS *less* THAN RIV16-C1); ▲ ENTIRELY ABOVE. RESIDENCE IS A *log loss* DIFFERENCE, SO A POSITIVE VALUE MEANS THE ARM SUPPLIES *less* RESIDENCE CAPABILITY. ACS 2018 DEVELOPMENT POOLS; THREE SEEDS. THE LAST COLUMN COUNTS SEEDS CLEARING THE LEGACY SERVICE-PROBE ALLOWANCE, OUT OF 3 (RIV16-C1 CLEARS 1/3).

Study 6 pilot: capability and disclosure did not separate (β increases to the left)

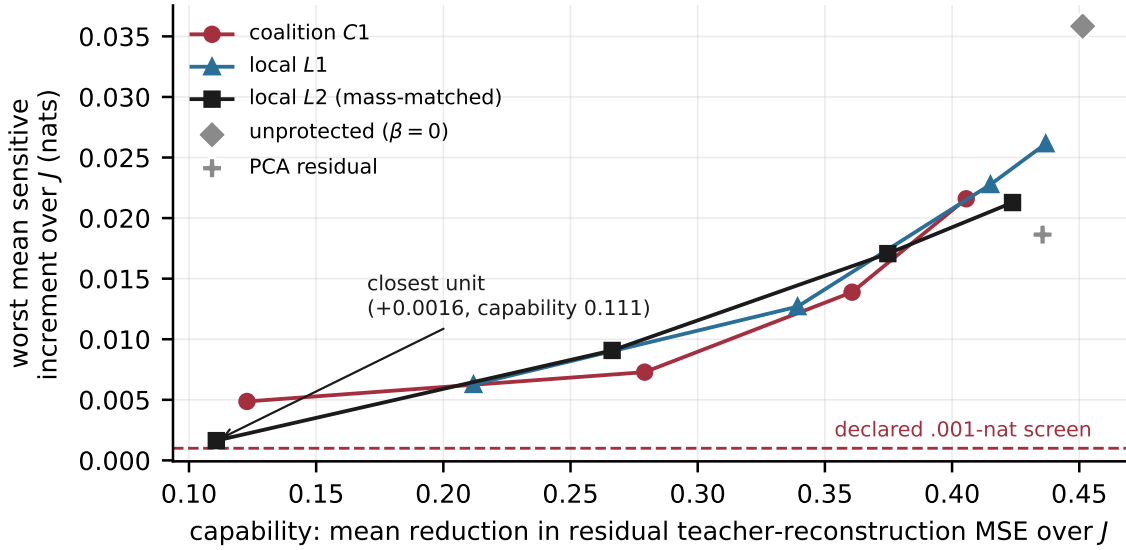


Fig. 3. Utility-first pilot (development pools): capability against the worst mean sensitive increment over J , with the declared screen dashed. Increasing the adversarial weight moves each policy down and to the left. Point-estimate screening values, not confidence statements.

“no extension of J in this family separates them”. This design cannot distinguish them.

VII. A CAPACITY DIAGNOSTIC, AND WHAT IT REJECTS

The most recent study asked whether a *randomized* finite release, optimised under explicit conditional-disclosure constraints, could reach an operating point the deterministic extension could not. It registered a tier ladder with machine-readable gates and stopped at the first empirical gate.

The stochastic solver was never run on data. The finite-channel machinery was built and validated on synthetic fixtures—where randomization does strictly beat the best deterministic perfectly-private map—but no channel was ever constrained, fitted or audited on survey data, and no cloud

compute was spent. **This is therefore not a failed stochastic mechanism**, and nothing here bears on whether a role-constrained stochastic release would beat a deterministic one. What failed is a prior capacity question.

What was measured. A prespecified, label-free k -means code T of the permitted A-side inference inputs, at 64 states and at one predeclared 256-state fallback, was given to the house utility probe family alongside the same-host J release, and asked whether it retained a residence advantage *before* any privacy constraint. It did not: the anchor-mean advantage is -0.00268 nats at 64 codes and -0.01136 at 256, with no anchor positive under the unweighted accounting (Table II, Fig. 4). The gate failed and stages C and D did not run.

The informative part is why. The code is not noise: alone

code size	anchor	unweighted	person-weighted	paired SE
$ T = 64$	0	-0.00091	+0.00216	0.00553
	1	-0.00472	-0.00400	0.00479
	2	-0.00241	-0.00682	0.00587
anchor mean		-0.00268	-0.00288	
$ T = 256$	0	-0.00404	+0.00007	0.00564
	1	-0.01710	-0.01315	0.00557
	2	-0.01295	-0.01509	0.00491
anchor mean		-0.01136	-0.00939	

TABLE II

CAPACITY DIAGNOSTIC, DEVELOPMENT POOLS, VALIDATION SPLIT: RESIDENCE ADVANTAGE OF $[H_A, Z_J, \text{onehot}(T)]$ OVER SAME-HOST J , PER ANCHOR AND WEIGHTING, WITH THE PAIRED HOUSEHOLD-BOOTSTRAP STANDARD ERROR. AT $|T| = 64$ EVERY ESTIMATE LIES WITHIN ABOUT ONE SE OF ZERO: NO DEMONSTRATED ADVANTAGE, AND NO DEMONSTRATED DEFICIT.

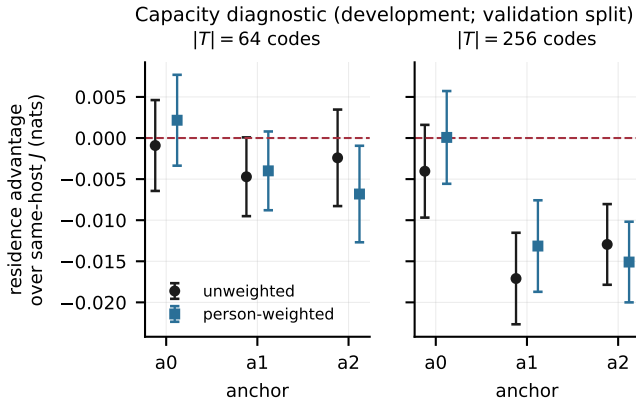


Fig. 4. The same diagnostic with paired household-bootstrap standard errors. Development evidence on repeatedly used pools; the test split was never read.

it beats the label prior by 0.014–0.022 nats in every anchor. Alone it is also beaten by J in every anchor, and adding it to J changes nothing measurable (Table III). The plain reading is that the A-side residence signal this code captures is, at this probe family’s resolution, *already carried by the service predictions that were published*. That is a statement about the release contract, and it offers a mechanism-level explanation for why this line of work has repeatedly gone negative: the contract already hands the recipient the signal an extension is trying to add.

A. What this diagnostic does not establish

Four delimitations, each of which we enforce in the wording above.

It is a fitted-probe result, not an information statement.

The differences that motivate it are 0.16, 0.98 and 0.41 standard errors from zero at the declared resolution, on 2048 fitting rows with 84 input columns against the baseline’s 20, using a logistic and a 40-epoch MLP selected on validation log loss. “No measurable improvement at this probe capacity” is supported; “ T contains no useful information beyond J ” is not. A probe family can miss information its own inputs jointly determine: in an exact fixture with $Y = J \cdot T$ for independent

anchor	prior [†]	T only [†]	J only	$J + T$
0	0.54501	0.53093	0.52507	0.52598
1	0.52859	0.51292	0.50114	0.50586
2	0.55554	0.53321	0.52856	0.53097

TABLE III

RESIDENCE VALIDATION LOG LOSS (NATS, LOWER IS BETTER) AT $|T| = 64$.

[†]THE PRIOR AND T -ONLY COLUMNS ARE REPORTED BY THE SOURCE STUDY BUT HAVE NO MACHINE-READABLE BACKING OR COMMITTED GENERATOR AT THE PINNED COMMIT; WE REPORT THEM AS REPORTED-NOT-VERIFIED. THE J AND $J+T$ COLUMNS REPRODUCE EXACTLY FROM ITS STORED RESULTS.

signs, a score additive in J and one-hot T gains exactly zero over the prior while the joint view determines Y . That is a capacity counterexample, not a claim that residence behaves like this.

It rejects appending, not replacing. The diagnostic asks whether T adds to J . It cannot see a release that *replaces* J . In an exact fixture with H constant, $J = (Y, S)$ and $T = Y$ for independent bits, appending T to J adds exactly zero task information, while replacing J by T preserves task utility in full and removes the sensitive bit entirely (0.693 nats of disclosure removed). Append-redundancy is therefore not a universal rejection of replacement channels, and the untested question—a prospective release beside unchanged H for a new recipient, with J as a comparator rather than a component—remains open.

It is one code, one probe family, two resolutions, three anchors, on repeatedly used development pools. It is not a capacity theorem and not a statement about the population. Higher resolution is monotonically worse, which is consistent both with there being nothing to find and with the probe being handicapped by width.

One reported ceiling is withdrawn. The study also estimated a fitted-model ceiling under a partition it described as “baseline probability deciles”. Those deciles are constructed from the baseline probe’s *per-row log loss*, which is computed from each person’s own residence label, so the partition conditions on the outcome. The signature is visible in the numbers: ten cells take residence log loss from about 0.52 to 0.065–0.094 nats. A fixture reproduces the effect on a label that is pure noise at the same base rate: loss-decile conditioning “predicts” it at 0.057 nats against a true entropy of 0.590, while the label-free control (deciles of the predicted probability) leaks nothing. **We withdraw that estimate rather than reinterpret it**, and we use no ceiling in any claim here. Its companion partition is label-free but has 1.9–2.1 rows per cell and is unusable for a different reason. Neither entered the gate decision, which rests on the deployable probe comparison alone; we therefore do not propagate the defect to the gate outcome or to any other result.

VIII. THE RELEASE CONTRACT, AND ONE ASSUMPTION WE COULD NOT VERIFY

The interface of §II says H_A and H_B are immutable published outputs. J is different: it is an experimental channel

produced inside this project. In the source studies, “released wire” denotes the experimental release interface—arrays written to disk and audited—not delivery to an external recipient, and we found no evidence that J was ever deployed to anyone.

This matters because the most recent design treated J as a compulsory component of the new release, on the stated ground that it “was actually released to these recipients and remains accessible” and so could not be revoked. We could not substantiate that, and a later registration asserting prior deployment is not evidence of deployment. The consequence is structural rather than cosmetic: holding J inside the release turned the capacity question into the append question of §VII, which is precisely the question that cannot see a replacement win. **If the channel was never deployed, then J is a comparator and not a compulsory component**, and a release for a *new* recipient under a prospective contract is the natural next design. If deployment evidence exists, the old-recipient threat model returns and must be audited as such; we state the dependency rather than resolve it by assumption.

IX. WHAT THIS EVIDENCE DOES NOT ESTABLISH

No attack maximum is an upper bound. Every recovery number is what a validation-selected member of a finite collection achieved. Three separate searches for a stronger attacker—a wider, deeper network held out of every training ensemble, and two calibration suites—each failed to beat the standard slate, which bounds what those searches found, not what exists.

Development against confirmation. Only the locked-year transport is confirmatory. The 2018 pools have been used repeatedly, and three anchors with simultaneous intervals do not restore pristine confirmation. A later survey year remains sealed and unscored: it has never been loaded, transformed, inspected, scored or included in any restore bundle.

Screening thresholds are not confidence statements. Where a study nominated nothing, no adjusted interval exists, and its pass/fail legs are point estimates.

Precision near equality is limited. On these endpoints and pools, an adjusted one-sided upper bound at or below 0.001 nats was not achievable for a comparison whose point estimate sat near zero: the smallest adjusted half-width in the record is 0.00220 nats and the medians run 0.0069–0.0136. This already holds unadjusted, so multiplicity is not the cause. It is a statement about *those comparisons in the near-equality regime*, not about the achievable precision of a mechanism that does not yet exist: the standard error belongs to the paired per-household sensitive losses of a specific comparison, and a release whose sensitive predictions nearly match the reference can have a much smaller one while its task utility differs substantially.

Audits do not transport across hardware below about 0.003 nats, because re-auditing refits the attacker slate; comparisons are therefore made against same-host references and not differenced against earlier published audits at finer precision.

Machinery is itself an artifact. One study’s audit slate omitted the in-slate untouched- J predictor its method text promised,

so a safeguard was served by a differenced comparator instead; the fix can only raise measured increments and therefore makes screens harder. Another study’s training-time penalty registers a positive gain for a provably non-disclosing constant channel, which affects the interpretation of its weight knob rather than its audited leakage; its magnitude in that study is unmeasured rather than estimated.

Nothing here is a certificate. Vanishing fitted moments imply neither conditional independence nor any bound on $I(S; Z \mid H)$, and a guarantee in a fitted finite model is not a population guarantee: in a validated fixture, binning a conditioning view drops a full log2 of measured leakage to exactly zero.

X. OPEN SCIENCE

Every number in this paper is generated from committed evidence at pinned commits, and each printed digit traces to a (commit, path, SHA-256) triple recorded in the submitted artifact. Table and figure generators, the independent verification scripts, the claim ledger, the correction log and the mathematical fixtures are included; no number was typed by hand. The evidence is ACS public-use microdata, so no restricted data is redistributed. Anonymised artifacts will be shared within the required window, and the final artifact deposited on acceptance.

XI. LLM USAGE CONSIDERATIONS

This work was substantially assisted by a large language model, and describing that as editorial help would be inaccurate. A model was used to write and refactor experiment and analysis code, to draft and revise this manuscript, to audit the studies’ own reports against their source code and stored outputs, and to construct the counterexample fixtures cited here. Experimental designs, registrations and stopping rules were fixed in advance of outcomes and are committed; model-generated analyses were checked by independently written scripts that recompute each reported quantity from the smallest machine-readable source, and several claims in this paper exist *because* that recomputation contradicted the original write-up—the withdrawn ceiling of §VII-A is one. The authors take full responsibility for every claim, number and omission in the submitted work, including those first drafted by a model; nothing was accepted on the model’s assertion alone, and every load-bearing figure is reproducible from the artifact.

XII. CONCLUSION

A service that has already published predictions can add a reusable channel beside them without changing a released number, and can steer that channel away from directions a recipient or a colluding pair would use to infer sex and race. On a locked, previously unused year, coalition conditioning does what it claims against both matched local controls, at a utility cost the data cannot bound below the margin its rule used. Everything we built afterwards to make that into a competitive release failed, in both registered senses of competitiveness, and

adapted published erasers applied to a channel we already had beat our own mechanism.

The most recent diagnostic suggests a reason that is specific to this contract rather than to any one optimiser: a label-free code of the permitted inputs carries real signal for the held-out task, and adds nothing measurable once the published service predictions are already in the recipient’s view. If that holds more broadly, then the room for an *additional* channel beside a rich published output is small, and the open question is not a better penalty but whether a release that *replaces* rather than *appends* can hold task utility while removing disclosure—which our evidence does not test.

The defensible contribution is the release problem and the evaluation design: fixed published outputs as a hard constraint, recipient-specific and coalition-specific audit roles, fresh and frozen attackers never pooled, absolute and incremental recovery side by side, negative increments unclipped, denominators beside every count, per-endpoint results never averaged, and a sealed temporal transport with its multiplicity fixed in advance. That contract is what made six negative results legible instead of ambiguous, and it is what we would keep.

REFERENCES

- [1] D. Madras, E. Creager, T. Pitassi, and R. Zemel, “Learning adversarially fair and transferable representations,” in *Proceedings of the 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 80. PMLR, 2018, pp. 3384–3393. [Online]. Available: <https://proceedings.mlr.press/v80/madras18a.html>
- [2] D. McAllester and K. Stratos, “Formal limitations on the measurement of mutual information,” in *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 108. PMLR, 2020, pp. 875–884, arXiv:1811.04251. [Online]. Available: <https://proceedings.mlr.press/v108/mcallester20a.html>
- [3] K. Fan, “On a theorem of Weyl concerning eigenvalues of linear transformations I,” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 35, no. 11, pp. 652–655, 1949.
- [4] K. Zhang, J. Peters, D. Janzing, and B. Schölkopf, “Kernel-based conditional independence test and application in causal discovery,” in *Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2011, pp. 804–813, arXiv:1202.3775. [Online]. Available: <https://arxiv.org/abs/1202.3775>
- [5] E. V. Strobl, K. Zhang, and S. Visweswaran, “Approximate kernel-based conditional independence tests for fast non-parametric causal discovery,” *Journal of Causal Inference*, vol. 7, no. 1, 2019, arXiv:1702.03877. [Online]. Available: <https://arxiv.org/abs/1702.03877>
- [6] B. Sadeghi, R. Yu, and V. N. Boddeti, “On the global optima of kernelized adversarial representation learning,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, arXiv:1910.07423. [Online]. Available: <https://arxiv.org/abs/1910.07423>
- [7] B. Sadeghi, L. Wang, and V. N. Boddeti, “Adversarial representation learning with closed-form solvers,” in *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2021)*, 2021, pp. 731–748, arXiv:2109.05535; pages and DOI from dblp/search listing (Springer page not loadable). [Online]. Available: <https://arxiv.org/abs/2109.05535>
- [8] B. Sadeghi, S. Dehdashtian, and V. N. Boddeti, “On characterizing the trade-off in invariant representation learning,” *Transactions on Machine Learning Research*, 2022, arXiv:2109.03386; OpenReview id 3gfpBR1ncr. [Online]. Available: <https://arxiv.org/abs/2109.03386>
- [9] F. Ding, M. Hardt, J. P. Miller, and L. Schmidt, “Retiring adult: New datasets for fair machine learning,” in *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- [10] S. Dehdashtian, B. Sadeghi, and V. N. Boddeti, “Utility-fairness trade-offs and how to find them,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, arXiv:2404.09454. [Online]. Available: <https://arxiv.org/abs/2404.09454>
- [11] N. Belrose, D. Schneider-Joseph, S. Ravfogel, R. Cotterell, E. Raff, and S. Biderman, “LEACE: Perfect linear concept erasure in closed form,” in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023, arXiv:2306.03819. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/d066d21c619d0a78c5b557fa3291a8f4-Abstract-Conference.html
- [12] F. Holstege, S. Ravfogel, and B. Wouters, “Preserving task-relevant information under linear concept removal,” in *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025, arXiv:2506.10703. [Online]. Available: <https://arxiv.org/abs/2506.10703>
- [13] Y. Elazar and Y. Goldberg, “Adversarial removal of demographic attributes from text data,” in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. Brussels, Belgium: Association for Computational Linguistics, 2018, pp. 11–21. [Online]. Available: <https://aclanthology.org/D18-1002/>
- [14] T. Stadler, B. Kulynych, M. Gastpar, N. Papernot, and C. Troncoso, “The fundamental limits of least-privilege learning,” in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 46 393–46 411. [Online]. Available: <https://proceedings.mlr.press/v235/stadler24a.html>
- [15] L. Sankar, S. R. Rajagopalan, and H. V. Poor, “Utility-privacy tradeoffs in databases: An information-theoretic approach,” *IEEE Transactions on Information Forensics and Security*, 2013, arXiv:1102.3751. [Online]. Available: <https://arxiv.org/abs/1102.3751>

APPENDIX A STUDY CHRONOLOGY

APPENDIX B CLAIM LEDGER SUMMARY

APPENDIX C ADDITIONAL TABLES

APPENDIX D CORRECTIONS THIS REVISION MAKES

Each item names the original claim, the evidence, and the replacement.

- 1) **“The inseparability appeared on a proxy that never saw a residential label, which removes the utility-target explanation.”** Withdrawn: the proxy is the residual of the unprotected channel, so proxy gains can be bought with sensitive-correlated directions (§VI-E).
- 2) **A fitted-model ceiling from “probability deciles”.** Withdrawn: the deciles are built from per-row log loss, which reads the person’s label (§VII-A). No ceiling is used.
- 3) **“The code’s residence content is already subsumed by J .”** Replaced by the measured fitted-probe result; the differences are 0.16–0.98 SE (§VII-A).
- 4) **“No anchor positive at either resolution.”** True unweighted; person-weighted, one of three anchors is positive at both resolutions. Anchor means stay negative and the gate outcome is unchanged.
- 5) **“A .001-nat confirmation is not demonstrable for any mechanism.”** Withdrawn by the source study and by us: the standard error belongs to a specific comparison, not to the pool (§IX).
- 6) **“A confirmable pass requires a candidate that reduces measured sensitive recovery.”** Scoped: valid only where the release replaces J ; for an appended release the

study	construction	evidence	status	what it established
1	locked ACS 2017 transport	349efa45	confirmatory	coalition effect over both local controls; the .001 point rule, not interval noninferiority
2	nonlinear penalty / rank	c37807e4	development	negative; penalty coordinate-dependent, diagnosed after evaluation
3	invariant repair + external baselines	73903b7f	development	negative; closed-form erasers beat the developed mechanism
4	direct adversarial refinement from A_0	69e790af	development	negative on both years
5	J -initialised factorial + coalition projection	a56bcc7f	development	negative; no prospectively selected release met the conjunction
6	utility-first residual extension of J	ba531ab4	development	negative pilot: capability gained on the proxy, disclosure screen failed at every configuration; expansion not triggered

TABLE IV

THE COMPLETED STUDIES, WITH EVIDENCE COMMITS AND WHAT EACH IS EVIDENCE FOR. ONLY THE FIRST IS CONFIRMATORY.

inclusion fact makes the population increment non-negative, so a strictly negative requirement would certify finite-probe noise (§III, §VIII).

- 7) “ **J was already released to these recipients.**” Not substantiated; treated as an internal comparator (§VIII).
- 8) “**A release close enough to J to be confirmable would be uninteresting.**” Withdrawn: low variance on sensitive endpoints is compatible with a large task-utility difference (§IX).

Claim	Status	Supporting measurement	What it does not establish
H preserved exactly in every release	Structural identity	Verified bitwise, 210/210 views	Says nothing about service <i>accuracy</i> , which moved by up to 0.015 nats
Appended channel adds authorised capability	Empirical, development & sealed	0.017–0.032 nats residence gain over H	Half-headroom reference fails in 1–3 of 3 seeds for every arm
Coalition penalty beats both local controls	Confirmatory (2017 sealed)	14/16 cells better, 0 worse, 2 unresolved	One year, one state, three seeds; residence cost not bounded below 0.001
Developed mechanism vs. frozen neural channel	Empirical, development	Worse on 3 of 4 sensitive endpoints	J is internal, and it supplies <i>less</i> residence capability; neither dominates
Rotation defect repaired exactly	Elementary algebra + numerics	Invariance to 10^{-15} , 18/18 cells	Invariance of an objective is not evidence the objective measures the right thing
Repair reduces measured disclosure	Refuted	8 of 48 cells significantly worse, 0 better	Attributable to the repaired <i>package</i> , not to any single changed ingredient
External adaptations beat the developed mechanism	Empirical, development	LEACE / SPLINCE / OptNet-C1 lower on all four endpoints	Adaptations, not replicas; LEACE costs residence against its own source channel
Any bound on $I(S; Z H)$	Not claimed	–	Finite validation-selected attacks attain no infimum in either direction
Study 5: competitive conjunction met by a panel release	Not established	0 of 4 nominees, both weightings, ref_0 or historical J	Not a proof no competitive release exists; one objective, one selection rule
J -initialised training improves on J	Not supported	Selection returned J (step 0) in every J -arm at $\gamma \leq .01$	Other selection yardsticks untested
Coalition projection on J beats matched local projections	Not established	No candidate-wide difference vs L , LX (family P)	Nonsignificance is not equivalence
Projection of J lowers coalition sex recovery	Empirical, development	AB/SEX -0.0091 vs J , residence loss $+0.0101$ (unweighted)	A trade-off, not dominance; ordinary LEACE on J is as close as ± 0.0011 (points)
Coalition race effect on A_0 at $k=2$	Exploratory (family X)	Below both local controls, both weightings	Not nominated; residence unresolved, not preserved; far from J
Stronger attack confirms protection	Not claimed	MLP[256,256,128] weaker than standard audit	Survival untested
Study 6: extension adds capability on the proxy	Empirical, development	0.111-0.451 reduction in residual reconstruction error; 14/14 configurations	Reconstruction is a proxy, not residence transfer; no residential label entered it
Study 6: added disclosure within the declared .001-nat screen	Not established	0 of 14 configurations; closest $+0.0016$ unw / $+0.0021$ pw	Point-estimate screen, no intervals (nothing nominated); one family, $r=2$
Study 6: capability and disclosure separable by the adversarial term	Not established	Both fall monotonically in β in all three policies	This finite family and budget only
Study 6: coalition benefit over matched local control	Not established	$C1$ $+0.0049$ vs $L2$ $+0.0016$ at $\beta=30$	Third mechanism family to reproduce this
Utility-first candidate hidden in the earlier 372-slot evidence	None	0 of 124 configurations pass vs J or $leace_A0$ ($m=2480$)	Exploratory reanalysis; does not change that study's own verdict
Study 6: stronger attacker found	Not established	0 of 8 prespecified recipes beat the standard slate	Not evidence of survival against a stronger attack
Study 6 Tier 3 / Tier 4 results	NOT TRIGGERED	Gate rule after the failed pilot; no partial grid fitted	Absent evidence for an unrun design, not missing evidence for a claimed one

TABLE V

EVERY LOAD-BEARING CLAIM, ITS STATUS, THE MEASUREMENT BEHIND IT, AND WHAT IT DOES NOT ESTABLISH. THE FULL TRACEABLE LIST IS IN THE ARTIFACT.

Arm – J	wt.	A / sex	AB / sex	A / race	AB / race	residence
LEACE	unw.	+0.0055	–0.0020	+0.0019	–0.0022	–0.0005
	PWGTP	+0.0056	–0.0016	+0.0034	–0.0027	–0.0010
SPLINCE	unw.	+0.0047	–0.0011	–0.0034	–0.0028	+0.0048
	PWGTP	+0.0052	–0.0007	–0.0013	–0.0025	+0.0041
OptNet-C1	unw.	+0.0061	–0.0044	–0.0012	–0.0009	+0.0045
	PWGTP	+0.0064	–0.0041	+0.0005	–0.0012	+0.0026
OptNet-L1	unw.	+0.0104▲	+0.0017	+0.0093	+0.0077	+0.0017
	PWGTP	+0.0094▲	+0.0008	+0.0073	+0.0064	–0.0004
OptNet-L2	unw.	+0.0031	–0.0026	–0.0035	–0.0062	+0.0056
	PWGTP	+0.0028	–0.0035	–0.0022	–0.0073	+0.0032
riv16-C1	unw.	+0.0135▲	+0.0066	+0.0149▲	+0.0136▲	–0.0041
	PWGTP	+0.0152▲	+0.0077	+0.0188▲	+0.0139▲	–0.0050
riv8-C1	unw.	+0.0121▲	+0.0058	+0.0063	+0.0049	–0.0049
	PWGTP	+0.0147▲	+0.0070▲	+0.0099	+0.0078	–0.0052

TABLE VI

EXTERNAL ARMS AGAINST J . A BLANK CELL IS AN UNRESOLVED DIFFERENCE AT THREE SEEDS, NOT A DEMONSTRATED EQUALITY; THE ADJUSTED INTERVAL CONTAINS ZERO AND ALSO CONTAINS EFFECTS OF THE SIZE THIS TABLE REPORTS ELSEWHERE. NO EQUIVALENCE OR NON-INFERIORITY TEST WAS RUN FOR THESE CELLS. NEGATIVE RESIDENCE MEANS THE ARM SUPPLIES *more* RESIDENCE CAPABILITY THAN J .

Quantity	Defective objective (Study 2)	Repaired objective (Study 3)
Rotation-only share of training gain, attained (<i>a property of the objective</i>)	mean 0.628, range 0.367–0.907 over 18 cells	max. abs. 1.4×10^{-15} over 18 cells
Direct invariance identity $ L(WQ) - L(W) $ on real data	up to 7.5×10^{-3}	5.6×10^{-17}
<i>Measured disclosure, repaired minus defective, over 6 matched contrasts $\times$ 4 sensitive endpoints $\times$ 2 weightings:</i>		
Sensitive cells significantly worse (more recovery)	8 of 48	
Sensitive cells significantly better (less recovery)	0 of 48	
Largest single increase in recovery	+0.0107 A / race, riv16-L2	
Residence cells resolved in either direction	0 of 12	

TABLE VII

THE EXACT REPAIR WORKED ON THE OBJECTIVE AND DID NOT WORK ON THE RELEASE. THE UPPER ROWS ARE A DIAGNOSTIC OF THE *objective*; THE LOWER BLOCK IS A MEASUREMENT OF THE *release*. THEY ARE REPORTED SEPARATELY BECAUSE AN EXACTLY INVARIANT OBJECTIVE DID NOT PRODUCE LESS MEASURED DISCLOSURE. THE ATTAINED ROTATION SHARE IS NOT A SUPREMUM – EVERY ROTATION SEARCH STOPPED ON BUDGET EXHAUSTION OR A FAILED LINE SEARCH – AND IT IS NOT A CAUSAL DECOMPOSITION OF WHAT THE OPTIMISER DID.

tier	status	cost	outcome
T0 infrastructure, archive, restore	PASS	\$0.16	bytes restored and re-hashed; frozen mappers re-execute
T1 evidence reuse + audit calibration	PASS	\$0.20	0 of 124 stored configurations meet the utility-first criterion; 0 of 8 attacker recipes stronger than the standard slate
T2 pilot (42 units)	FAIL	\$0.35	capability leg passed for all 14 configurations; the .001-nat disclosure screen failed for all 14
T3 expansion (93 further slots)	NOT TRIGGERED	—	forbidden by the T2 gate rule; no partial grid was fitted
T4 nominee comparison, 2017 transport	NOT TRIGGERED	—	nothing was nominated

TABLE VIII

THE UTILITY-FIRST STUDY’S TIER LADDER. TIERS 3 AND 4 WERE NEVER TRIGGERED.